POLITECNICO DI MILANO
DEPARTMENT OF AEROSPACE ENGINEERING
AEROSPACE SYSTEM COURSE

PROVA FINALE: INTRODUZIONE ALL'ANALISI DI MISSIONI SPAZIALI

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Index

1 Introduction **2**

2 Initial and final orbit characterization **2**

2.1 Initial orbit 2

2.2 Final orbit 3

2.3 Representation 4

3 Transfer trajectory definition and analysis **4**

3.1 Standard strategy 4

3.2 Standard alternative 5

3.3 Direct maneuver 6

3.4 Bielliptical strategy 7

4 Conclusions **9**

5 Appendix **9**

5.1 Tables 9

1 Introduction

One of the most important skills for a space engineer is the ability to optimize a transfer from an initial orbit to a final orbit. The main goal of this project is to analyze and compare different transfer strategies, as well as discuss their pros and cons. The base case is a standard strategy, which means a transfer method only composed by maneuvers that change one aspect of the orbit at a time. This method will then be compared to other transfer strategies, both standard and not, intended to improve some aspects of the standard strategy, namely the required total impulse and the total time to transfer from the initial point to the final point. These strategies are going to be a method that involves only standard maneuvers; a direct transfer and a bielliptical transfer.

2 Initial and final orbit characterization

2.1 Initial orbit

2.1.1 Determine initial orbital parameters from given position and velocity

The initial position is given in the form of its position and velocity expressed in cartesian coordinates:

$$\vec{r}_1 = \begin{Bmatrix} -4944.4602 \\ 4989.7097 \\ 6462.6224 \end{Bmatrix} km \qquad \vec{v}_1 = \begin{Bmatrix} -4.7010 \\ -4.6070 \\ 0.3576 \end{Bmatrix} km/s$$

A MATLAB function was implemented to convert cartesian position and velocity into the corresponding keplerian parameters. When applied to the initial position and velocity it produced the following results:

a_1 (km)	e_1(-)	i_1 (rad)	Ω_1 (rad)	ω_1 (rad)	θ_1 (rad)
9948.348	0.057485	0.744467	0.834303	0.701428	0.830033

2.1.2 Discuss the result, evaluate other relevant orbit data

From these results, other orbital characteristics can be determined, such as the specific energy, the orbital period, the pericenter and apocenter radii and semilatus rectum:

$$\epsilon_1 = -\frac{\mu}{2*a_1} = -20.0335\ km^2\ s^{-2}; \qquad T_1 = 2\pi\sqrt{\frac{a_1^3}{\mu}} = 9875.0\ s;$$

$$r_{p1} = a_1 * (1 - e_1) = 9674.917\ km; \qquad r_{a1} = a_1 * (1 + e_1) = 10221.778\ km;$$

$$p_1 = a_1 * (1 - e_1^2) = 9915.473\ km;$$

2.2 Final orbit

2.2.1 Determine final orbit position and velocity at set point given keplerian parameters

The final point is given in terms of its keplerian parameters:

a_2 (km)	e_2(-)	i_2 (rad)	Ω_2 (rad)	ω_2 (rad)	θ_2 (rad)
14870.00	0.3004	0.9583	2.4930	2.9000	0.2117

A second MATLAB function has been implemented to output the corresponding cartesian position and velocity vectors, which are the following:

$$\vec{r}_2 = \begin{Bmatrix} 8221.2674 \\ -6457.0743 \\ 255.7253 \end{Bmatrix} km \qquad \vec{v}_2 = \begin{Bmatrix} 2.8743 \\ 2.8776 \\ -5.7348 \end{Bmatrix} km/s$$

2.2.2 Discuss the result, evaluate other relevant orbit data

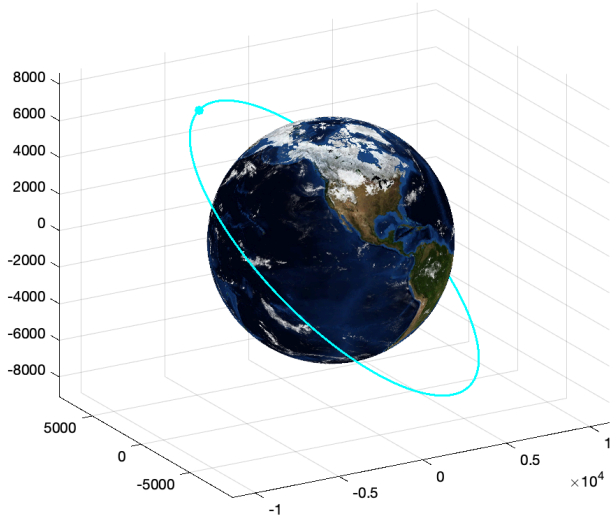
As for the initial orbit, the specific energy, the orbital period, the perigee and apogee radii and the semilatus rectum can be calculated from the keplerian parameters:

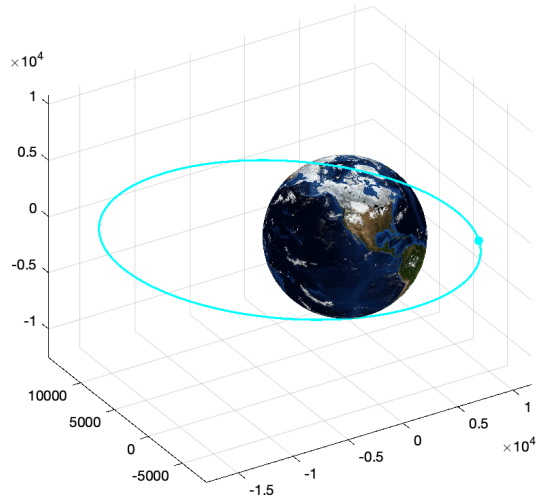
$$\epsilon_2 = -\frac{\mu}{2*a_2} = -13.4028\ km^2\ s^{-2}; \qquad p_2 = a_2 * (1 - e_2^2) = 10403.052\ km;$$

$$T_2 = 2\pi\sqrt{\frac{a_2^3}{\mu}} = 18045.8\ s; \quad r_{p2} = a_2 * (1 - e_2) = 10403.052\ km;$$

$$r_{a2} = a_2 * (1 + e_2) = 19336.948\ km;$$

2.3 Representation





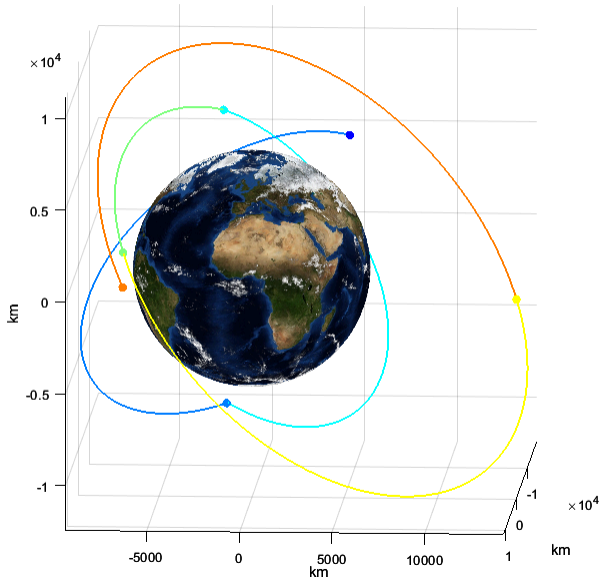
3 Transfer trajectory definition and analysis

3.1 Standard strategy

3.1.1 Characterization

The first strategy to be analyzed is the standard maneuver, which consists of a series of impulses organized in a specific order.
The first impulse involves a change of orbital plane, followed by a periapsis argument change, and finally a perigee-apogee transfer.

3.1.2 Execution



The standard maneuvers are applied in the chosen sequence; for the periapsis change the option that leads to the total minimum. The orbit's plane change is carried out through a single impulse maneuver. In this case the impulse is executed at the second intersection that the satellite crosses from its initial point because it results the most convenient in terms of speed variation and, consequently, cost.
The periapsis argument change is a maneuver involving the rotation of the eccentricity vector and is purely radial. This transfer can be performed at either of the two intersection points between the orbits, but the required impulse does not depend on the location where the maneuver is executed. The second point in the satellite's path has been chosen because it gives a smaller total travel time. Once the two orbits are aligned,it is possible to change the shape of the orbit with a perigee-apogee bi-tangent transfer. The satellite then coasts to the final point and the transfer is thus concluded.

3.1.3 Results

We use the data calculated with the standard maneuver to compare various strategies in order to optimize the assigned transfer as much as possible.

Δt_{tot} (s)	Δv_{tot} (km/s)
31661.1	8.7215

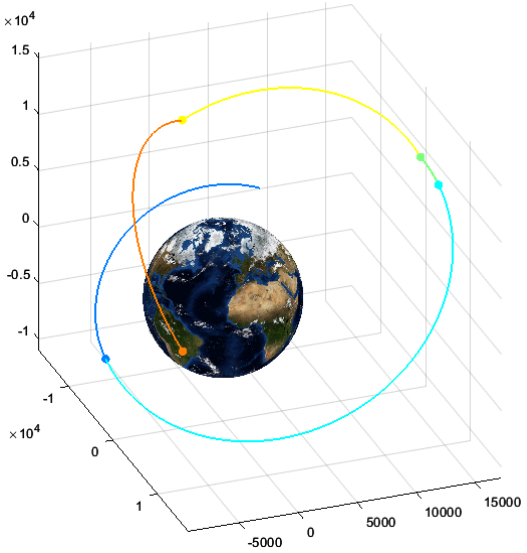
3.2 Standard alternative

3.2.1 Characterization

Searching for improvements on the standard strategy, the first option is using standard maneuvers in an order different from the previous one. To improve on the fuel cost, the plane change is by far the most expensive impulse, because it is performed on a small orbit, which translates to high transversal speed. By first enlarging the orbit, this cost can be reduced. For this reason, the present transfer strategy involves a shape change followed by a periapsis rotation and finally a plane change.

3.2.2 Execution

Since multiple points are available to perform the various maneuvers, all possible combinations were analyzed, and the most convenient in terms of Δv has been chosen to represent this transfer strategy.



3.2.3 Results

The most convenient combination of this strategy resulted in an apogee-apogee change of shape, followed by a periapsis rotation made at the intersection point with highest θ value, and a plane change executed again at the intersection point with highest θ value. The impulses and total time are listed in the following table:

Δt_{tot} (s)	Δv_{tot} (km/s)
23747.36	6.7227

3.3 Direct maneuver

3.3.1 Characterization

The fastest transfer method that transfers from one point to another is an orbit that passes through the two points. If the two points are intended as part of two orbits, as is the case in this study, this transfer method requires two impulses: the first allows the spacecraft to enter the transfer orbit, while the second puts the spacecraft onto the final orbit when the destination point is reached. Since both pulses involve a plane change, it is expected that this transfer strategy is going to be expensive in terms of the total Δv required. For this reason, this maneuver is intended for a military-like scenario, where the most important aspect is the reduction of transfer time, and the required fuel is less relevant.

3.3.2 Execution

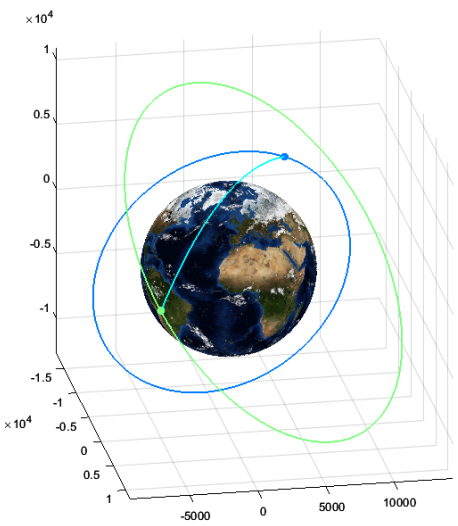
The transfer orbit plane is uniquely defined by three points: the starting position, the ending position and Earth's center. However, the shape of the orbit and the direction of travel are not uniquely defined. These are therefore going to be the parameters to be varied in an orbit optimisation algorithm where the goal is the smallest possible travel time. To define the direction of travel, both possible options are considered, and the shape of the orbit is determined by the argument of perigee.
The constraints are that the orbit must be elliptical and its perigee must be above Earth's atmosphere (150 km above Earth's surface has been chosen as the limit).

3.3.3 Results

Running the optimisation algorithm, the orbit with shortest travel time has the following parameters:

a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ_i (rad)	θ_f (rad)
$6.8741 \cdot 10^8$	0.999990	1.7012	2.4726	1.8664	5.3096	1.2505

As can be easily observed by the semi-major axis and eccentricity of the orbit, this type of transfer requires a very large and highly elliptical orbit, which is only used for a short section near the perigee.
In terms of time and Δv required, the results are in the following table:

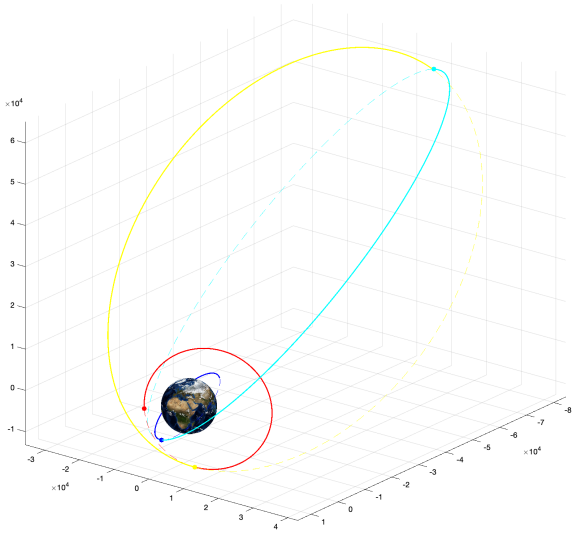


Δt (s)	Δv_{tot} (km/s)
1980.6	18.995395

This is clearly the fastest option, but requires a significant effort.

3.4 Bielliptical strategy

3.4.1 Characterization



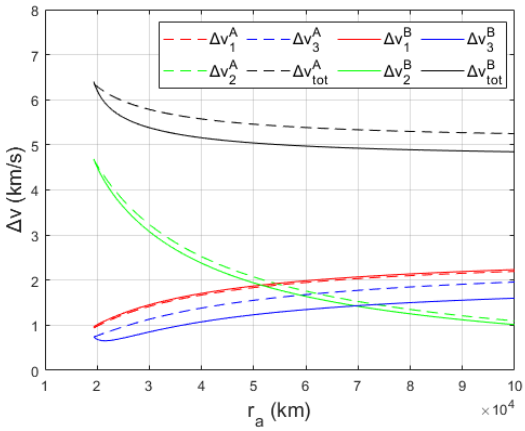
Because the initial and final orbits lie on different planes, at least one plane change is required to transfer from one orbit to the other. Since this is generally an expensive maneuver, to reduce its cost it should be performed where the transversal speed is low, which means as far away as possible. This can be done by first enlarging the orbit, then changing the orbital plane, and finally shrinking the orbit back to the final size. The first operation is done with a tangent impulse, which sends the spacecraft to the first transfer orbit. This orbit is coplanar and tangent to the initial orbit and has its apogee on the intersection of the initial and final planes. The spacecraft then coasts to the apogee of the transfer orbit, where it performs the second impulse. This sends the spacecraft on the second transfer orbit, which, similarly to the first one, is coplanar and tangent to the final orbit. The spacecraft coasts to the point of tangency of the two orbits, and performs the last burn to enter the final orbit. It then coasts to the final point.

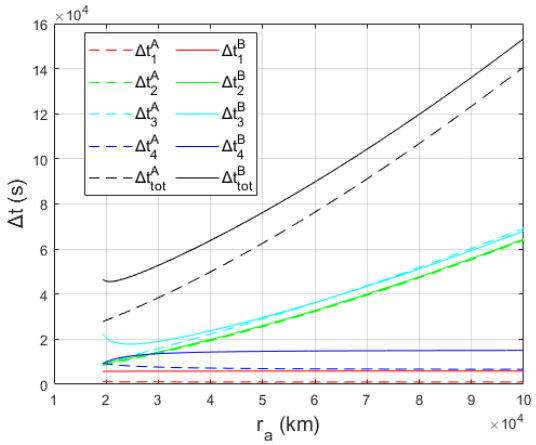
3.4.2 Execution

Since the only free parameter is the transfer orbits apogee, all other quantities are going to be functions of it. In principle, there is no limit on how far this point can be, but in reality there is such a limit because of the gravitational influence of other celestial bodies, (namely the Sun and the Moon). After some research on real satellites, an upper bound has been chosen to be 100 thousand kilometers from the center of the Earth.
To characterize the transfer orbits, a set of conditions is required: first, the radius at the point of tangency has to be the same; the difference in the angular position of the tangency point relative to the two perigees is the desidered periapsis rotation; the apogee radius of the transfer orbit has to equal the desidered radius; finally, the velocities on the two orbits need to be parallel. The last condition is translated in the ratio between the radial and transversal speeds, which has to be the same for both orbits. These are four equations with four unknowns; solving them gives the orbital parameters of the transfer orbit and the position of the point of tangency.

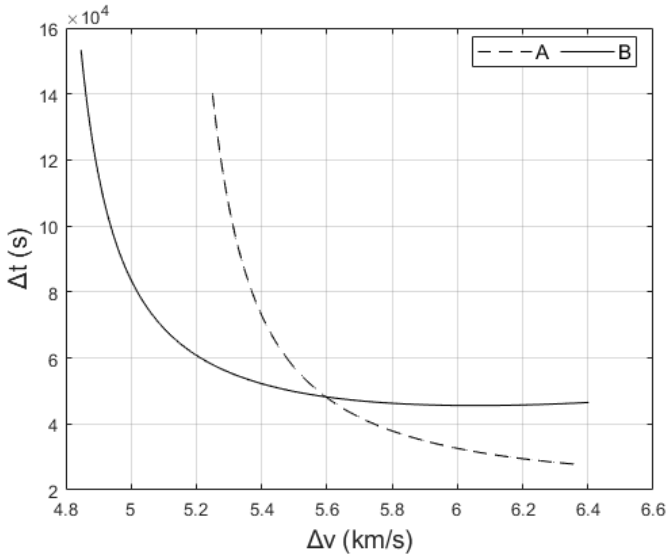
3.4.3 Results

Plotting the required impulses and times against the transfer orbits apogee gives the following graphs:





The A and B labels refer to the two directions in which it is possible to execute the maneuver. Plotting Δt against Δv results in the following curves:



As can be seen from these curves, increasing the apogee decreases the total cost of the maneuver, but increases the required time, both as expected. The reduced cost is due to the fact that this strategy performs multiple impulses at the same time: the first and third combine a shape change and a periapsis rotation, and the second one combines a plane change and a shape change. The increased time is due to the long periods of the transfer orbits: since they are very large they also have longer orbital periods.

Since the aim of this maneuver is the lowest possible Δv, the minimum is chosen, which gives the following results:

Δt (s)	Δv_{tot} (km/s)
153342.67	4.8466

4 Conclusions

The ones described above are just some of the possible transfer strategies: there are likely other types of transfer that optimize the cost or the time to connect the starting point to the finish point. Looking at the results, the direct maneuver can bring the spacecraft from the initial to the final point in the shortest amount of time, but this comes at a really high cost. The standard maneuver also has a high cost compared to the other analyzed strategies, and a time inferior only to the elliptical transfer. This means that it clearly is not the best option. The standard alternative is better both in terms of time and cost, but the bielliptical offers a lower cost. These two options are probably the best of the four proposed; the final choice depends on the importance given to the flight time and the fuel cost in the mission profile. In this analysis, the most important aspect has been chosen to be the transfer cost, therefore the best transfer strategy is the bielliptical transfer.

5 Appendix

5.1 Tables

Standard

	T[s]	a[km]	e	i [deg]	Ω [deg]	ω[deg]	θ[deg]	ΔV[km/s]
Initial orbit	0	9948.34	0.0575	42.66	47.80	40.19	47.55	0
Plane change	6088.8	9948.34	0.0575	42.66	47.80	40.19	258.30	7.0109
		9948.34	0.0575	54.91	142.84	328.40	258.30	
Apse line rotation	11232.6	9948.34	0.0575	54.91	142.84	328.40	98.87	0.7202
		9948.34	0.0575	54.91	142.84	166.16	261.12	
To first transfer orbit	13765.3	9948.34	0.0575	54.91	142.38	166.16	0	0.8620
		14356.7	0.3469	54.91	142.38	166.16	0	
To final transfer orbit	22325.1	14356.7	0.3469	54.91	142.38	166.16	180	0.1283
		14870.00	0.3004	54.91	142.38	166.16	180	
Final destination	31661.1	14870.00	0.3004	54.91	142.38	166.16	12.13	0
Total ΔV								8.7215

Standard strategy variant 1

	T[s]	a[km]	e	i [deg]	Ω [deg]	ω[deg]	θ[deg]	ΔV[km/s]
Initial orbit	0	9948.34	0.0575	42.66	47.80	40.19	47.55	0
circolurarization	3762.5	9948.34	0.0575	42.66	47.80	40.19	180	1.15027
		14870.00	0	42.66	47.80	0	0	
Plane change	14805.229	14870.00	0	42.66	47.80	0	78.30	5.7925
		14870.00	0	54.91	142.38	0	78.30	
Shape change	20793.104	14870.00	0	54.91	142.38	0	328.4	0.7635
		14870.00	0.3004	54.91	142.38	166.16	148.4	
Final destination	41259.4	14870.00	0.3004	54.91	142.38	166.16	12.13	0
Total ΔV								7.7063

Standard strategy variant 2

	T[s]	a[km]	e	i [deg]	Ω [deg]	ω[deg]	θ[deg]	ΔV[km/s]
Initial point	0	9948.34	0.0575	42.66	47.80	40.19	47.55	0
Shape change first point	3762.53	9948.34	0.0575	42.66	47.80	40.19	180	1.0297
		14928.60	0.2953	42.66	47.80	220.19	0	
Shape change second point	12838.85	14928.60	0.2953	42.66	47.80	220.19	180	0.0138
		14870.00	0.3004	42.66	47.80	220.19	180	
Periapsis change	13625.10	14870.00	0.3004	42.66	47.80	220.19	188.88	0.05033
		14870.00	0.3004	42.66	47.80	237.94	171.12	
Plane change	19113.34	14870.00	0.3004	42.66	47.80	237.94	240.55	5.1759
		14870.00	0.3004	54.91	142.84	166.16	240.55	
Final point	23747.36	14870.00	0.3004	54.91	142.84	166.16	12.13	0
Total ΔV								6.7227

Direct strategy 1

	T[s]	a[km]	e	i [deg]	Ω [deg]	ω[deg]	θ[deg]	ΔV[km/s]
Initial point	0	9948.34	0.0575	42.66	47.80	40.19	47.55	0
First impulse	0	9948.34	0.0575	42.66	47.80	40.19	47.55	11.9988
		$6.8641*10^8$	0.999990	97.47	141.67	106.94	304.22	
Second impulse	1980.6	$6.8641*10^8$	0.999990	97.47	141.67	106.94	71.65	6.9966
		14870.00	0.3004	54.91	142.84	166.16	12.13	
Final point	1980.6	14870.00	0.3004	54.91	142.84	166.16	12.13	0
Total ΔV								18.9954

Bielliptic-Bitangent strategy

	T[s]	a[km]	e	i [deg]	Ω [deg]	ω[deg]	θ[deg]	ΔV[km/s]
Initial point	0	9948.34	0.0575	42.66	47.80	40.19	47.55	0
First tangent point	5890.13	9948.34	0.0575	42.66	47.80	40.19	251.24	2.2331
		55033.90	0.8171	42.66	47.80	298.49	352.94	
Change of plane – change of shape	70279.86	55033.90	0.8171	42.66	47.80	298.49	180	1.0141
		56427.99	0.7722	54.91	142.84	226.70	180	
Second tangent point	138285.66	56427.99	0.7722	54.91	142.84	226.70	40.09	1.5994
		14870.00	0.3004	54.91	142.84	166.16	100.63	
Final point	153342.67	14870.00	0.3004	54.91	142.84	166.16	12.13	0
Total ΔV								4.8466

Direct strategy 2

	T[s]	a[km]	e	i [deg]	Ω [deg]	ω[deg]	θ[deg]	ΔV[km/s]
Initial orbit	0	9948.34	0.0575	42.66	47.80	40.19	47.55	0
First impulse	5521.43	9948.34	0.0575	42.66	47.80	40.19	238.38	0.8333
		13012.35	0.2165	42.25	52.92	285.04	349.75	
Second impulse	11166.03	13012.35	0.2165	42.25	52.92	285.04	190.19	5.0371
		14870.00	0.3004	54.91	142.38	166.16	241.86	
Final destination	15722.15	14870.00	0.3004	54.91	142.38	166.16	12.13	0
Total ΔV								5.8704